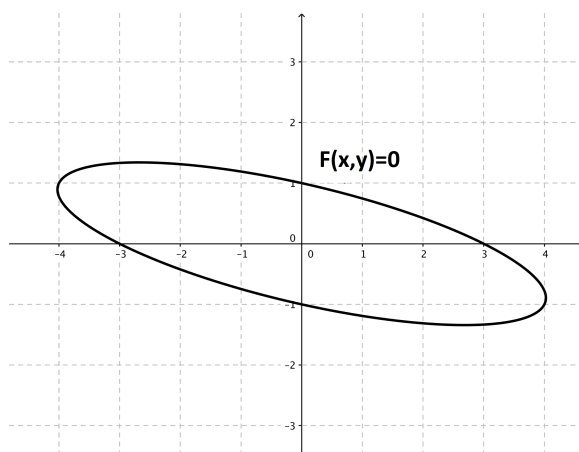


Problem 2

Problem 2

Problem 1 Consider the equation $x^2 + 4xy + 9y^2 = 9$. Note: This equation is equivalent to $x^2 + 4xy + 9y^2 - 9 = 0$. Therefore it has a form $F(x, y) = 0$



- Find $\frac{dy}{dx}$.
- Find the equation(s) of the tangent line(s) when $x = 0$. Draw the tangent line(s) on the above picture.
- Find the point(s) where the tangent line is horizontal. Draw the point(s) and line(s) on the above picture.